

Sample Paper 3 — Class 5 Mathematics Olympiad

Pattern: SOF IMO Level 1 — 50 questions, 60 marks, 60 minutes, no negative marking. Set a timer before you start.

Section 1 — Logical Reasoning (Q1–Q15, 1 mark each)

- Q1.** Find the odd one out: A) Hammer B) Saw C) Nail D) Shirt
- Q2.** Pilot is to Airplane as Captain is to _____. A) Ship B) Car C) Train D) Bicycle
- Q3.** If each letter of a word is moved 4 steps forward in the alphabet, LIME becomes PMQI. Using the same rule, how is ROSE written? A) VSWI B) VSWH C) USWI D) VTWI
- Q4.** Complete the series: 7, 14, 28, 56, ____ A) 98 B) 104 C) 112 D) 120
- Q5.** In a row of 32 students, Simran is 14th from the right. What is her position from the left? A) 18th B) 19th C) 20th D) 17th
- Q6.** Facing East, Priya turns 90° anticlockwise, then another 90° anticlockwise. Which direction does she face now? A) North B) South C) West D) East
- Q7.** Arrange in a logical order of education level: 1. University 2. Primary School 3. High School 4. Kindergarten A) 4, 2, 3, 1 B) 1, 2, 3, 4 C) 4, 3, 2, 1 D) 2, 4, 3, 1
- Q8.** Find the odd one out: A) 12 B) 18 C) 23 D) 30
- Q9.** If today is Friday, what day will it be after 31 days? A) Sunday B) Monday C) Tuesday D) Saturday
- Q10.** Sonia introduced a boy as the son of her father's brother. How is the boy related to Sonia? A) Brother B) Cousin C) Nephew D) Uncle
- Q11.** Find the missing number: 4, 9, 19, 39, 79, ____ A) 149 B) 154 C) 159 D) 164
- Q12.** In a certain code, WATER is written as YCVGT (each letter moved 2 steps forward). How is MONEY written in the same code? A) OQPGA B) OQPGZ C) OPPGA D) OQOGA
- Q13.** Find the odd one out: A) Cube B) Cuboid C) Square D) Pyramid
- Q14.** Neha scored more than Kabir. Kabir scored more than Meera. Arjun scored less than Meera. Who scored the lowest? A) Neha B) Kabir C) Meera D) Arjun
- Q15.** Complete the series: 1, 8, 27, 64, ____ A) 100 B) 110 C) 125 D) 150

Section 2 — Mathematical Reasoning (Q16–Q35, 1 mark each)

- Q16.** $67,845 + 28,956 = ?$ A) 96,801 B) 96,701 C) 95,801 D) 96,901
- Q17.** $70,002 - 38,459 = ?$ A) 31,543 B) 31,443 C) 31,643 D) 32,543
- Q18.** $6,789 \times 7 = ?$ A) 47,523 B) 47,423 C) 47,623 D) 46,523
- Q19.** $75,348 \div 6 = ?$ A) 12,558 B) 12,548 C) 12,568 D) 12,658
- Q20.** Find the HCF of 18 and 27. A) 6 B) 9 C) 3 D) 18
- Q21.** Find the LCM of 8 and 10. A) 18 B) 40 C) 80 D) 20
- Q22.** Which fraction is equivalent to $\frac{5}{8}$? A) $\frac{10}{16}$ B) $\frac{8}{5}$ C) $\frac{5}{16}$ D) $\frac{10}{8}$
- Q23.** $\frac{5}{12} + \frac{3}{12} + \frac{2}{12} = ?$ A) $\frac{5}{6}$ B) $\frac{10}{144}$ C) $\frac{5}{12}$ D) 1
- Q24.** Convert 0.28 to a fraction in simplest form. A) $\frac{7}{25}$ B) $\frac{7}{20}$ C) $\frac{2}{7}$ D) $\frac{14}{25}$
- Q25.** What is 65% of 180? A) 107 B) 112 C) 117 D) 122
- Q26.** Express $\frac{9}{20}$ as a percentage. A) 40% B) 45% C) 50% D) 55%
- Q27.** Two numbers are in the ratio 5:8. If the larger is 40, find the smaller. A) 20 B) 22 C) 25 D) 28
- Q28.** The cost price of an item is ₹540 and the selling price is ₹486. Find the loss percentage. A) 8% B) 10% C) 12% D) 15%
- Q29.** Find the simple interest on ₹6,000 at 9% per annum for 2 years. A) ₹1,000 B) ₹1,040 C) ₹1,080 D) ₹1,100
- Q30.** Convert 3 L 120 mL into millilitres. A) 3,120 B) 3,012 C) 3,102 D) 31,200
- Q31.** Convert 195 minutes into hours and minutes. A) 3 h 5 min B) 3 h 15 min C) 3 h 25 min D) 3 h 35 min
- Q32.** Find the perimeter of a rectangle with length 27 cm and breadth 13 cm. A) 40 cm B) 80 cm C) 160 cm D) 351 cm
- Q33.** Find the area of a square with side 11 cm. A) 111 sq cm B) 121 sq cm C) 44 sq cm D) 131 sq cm
- Q34.** How many lines of symmetry does a regular hexagon have? A) 4 B) 5 C) 6 D) 8
- Q35.** How many right angles make one and a half full turns? A) 4 B) 5 C) 6 D) 8

Section 3 — Everyday Mathematics (Q36–Q45, 1 mark each)

- Q36.** A factory fills 1,850 bottles each day. How many bottles does it fill in 18 days? A) 31,300 B) 33,300 C) 35,300 D) 32,300
- Q37.** Kavya had ₹9,500. She spent ₹3,200 and then ₹2,150. How much money does she have left? A) ₹4,150 B) ₹4,250 C) ₹4,050 D) ₹4,350

- Q38.** A tank holds 720 L. If $\frac{3}{4}$ of it is used, how much water is left? A) 160 L B) 170 L C) 180 L D) 190 L
- Q39.** 24 kg of wheat costs ₹720. Find the cost of 1 kg. A) ₹25 B) ₹28 C) ₹30 D) ₹32
- Q40.** 11 rows of chairs have 27 chairs each. How many chairs are there in total? A) 287 B) 297 C) 307 D) 317
- Q41.** A man deposits ₹5,000 at 9% simple interest per year. How much interest will he earn in 3 years? A) ₹1,250 B) ₹1,300 C) ₹1,350 D) ₹1,400
- Q42.** A shopkeeper bought goods for ₹3,200 and sold them for ₹2,880. Find the loss percentage. A) 8% B) 10% C) 12% D) 15%
- Q43.** A car travels at 80 km/h. How far will it travel in 2 hours 45 minutes? A) 200 km B) 210 km C) 220 km D) 230 km
- Q44.** A recipe needs $\frac{5}{6}$ cup of milk. How much milk is needed for 12 such recipes? A) 8 cups B) 9 cups C) 10 cups D) 11 cups
- Q45.** In a survey of 320 students, 55% like artwork. How many students like artwork? A) 166 B) 171 C) 176 D) 181

Section 4 — Achievers Section (Q46–Q50, 3 marks each)

- Q46.** A number is increased by 30% to get 390. Find the original number. A) 280 B) 290 C) 300 D) 310
- Q47.** The sum of three consecutive odd numbers is 99. Find the middle number. A) 31 B) 33 C) 35 D) 37
- Q48.** The simple interest on a sum for 5 years at 6% per annum is ₹2,400. Find the sum. A) ₹7,000 B) ₹7,500 C) ₹8,000 D) ₹8,500
- Q49.** Raj had some money. He spent $\frac{2}{5}$ of it on rent and $\frac{1}{3}$ of the remainder on groceries. He is left with ₹1,200. How much money did he have at first? A) ₹2,500 B) ₹2,800 C) ₹3,000 D) ₹3,200
- Q50.** The ratio of the ages of two cousins is 4:7. After 6 years, the ratio becomes 5:8. Find the younger cousin's present age. A) 20 B) 22 C) 24 D) 26

Answer key: `sample-paper-03-answer-key.md`
